UTKARSH ARYAN MISHRA

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SUMMARY

Final-year B.Tech IT student with hands-on cloud and DevOps experience across two internships, provisioning and deploying production infrastructure on AWS EC2 and EKS using Terraform, Docker, Kubernetes, and GitOps (Argo CD). AWS Certified Cloud Practitioner with practical exposure to CI/CD (GitHub Actions), Infrastructure as Code, and observability (Prometheus, Grafana). Backend development background in Node.js, Express.js, and MongoDB adds full-stack context to infrastructure work. Seeking entry-level Cloud, DevOps, or SRE roles.

TECHNICAL SKILLS

Cloud & Infrastructure: AWS, EC2, EKS, S3, CloudFront, IAM, Terraform, Linux
Containers & Orchestration: Docker, Kubernetes, Helm, NGINX Ingress
DevOps & GitOps: GitHub Actions, Argo CD, CI/CD, Infrastructure as Code, Git, GitHub
Observability & Security: Prometheus, Grafana, Trivy, Kubernetes HPA, NetworkPolicies
Backend Development: Node.js, Express.js, REST APIs, MongoDB (Mongoose ODM), MySQL
Programming: Python, JavaScript, Java, C
Testing & Tools: Jest, Supertest, Postman

EXPERIENCE

Backend & DevOps Intern | UPPCL — Uttar Pradesh Power Corporation Ltd. Jun 2026 – Jul 2026
IT Division, Lucknow | Backend Development · Cloud Deployment · Docker · Terraform · Nginx

- Built the backend for the Daily Revenue Monitoring Portal (DRMP), aggregating daily revenue data from 3 payment streams (Bank Wire, Payment Gateway, Billing System) across UPPCL divisions; designed 25+ REST API endpoints with JWT auth, bcrypt hashing, RBAC, and server-side validation.
- Architected 4 MongoDB collections (Users, Payment Submissions, Consolidated Reports, Audit Logs) via Mongoose ODM with an asynchronous audit-trail logging service, prior-date continuity checks, and PDFKit/ExcelJS report exports.
- Tested 25+ REST endpoints with Jest and Supertest and validated integrations via Postman; deployed the application on AWS EC2 provisioned via Terraform, containerized with Docker (non-root, multi-stage builds), and configured Nginx reverse proxy with Certbot SSL and a GitHub Actions CI/CD pipeline.

Software Engineering Intern | i2k2 Networks Pvt. Ltd. Jan 2026 – Apr 2026
Noida, Uttar Pradesh | Cloud & Managed Services Company

- Managed version control and branching workflows using Git & GitHub for collaborative backend code across real-world SDLC processes.
- Applied CI/CD pipeline concepts, Docker containerization, and AWS cloud infrastructure patterns in a professional DevOps setting.

PROJECTS

Kubernetes Platform Engineering & GitOps Lab | Repo
Kubernetes · Helm · Argo CD · Terraform · Prometheus · Grafana · Trivy

- Architected modular Terraform IaC provisioning a multi-AZ AWS VPC (4 subnets, 2 AZs) and EKS v1.30 cluster, with Argo CD + Helm v3 automating GitOps delivery — achieving commit-to-production rollout in under 3 minutes with zero manual cluster access.
- Engineered hardened multi-stage Docker builds gated by Aqua Trivy CVE scanning, reducing image size by 79% (1.1GB → 227MB) and blocking any build containing High/Critical vulnerabilities from reaching the registry.
- Configured Kubernetes HPA (2→5 replicas at 60% CPU) and Pod Disruption Budgets (minAvailable: 2); verified via chaos testing to achieve sub-2-second failover recovery with zero dropped requests during simulated pod crashes.
- Instrumented Google SRE RED-methodology observability (rate, errors, duration) via Prometheus Operator ServiceMonitors at 15-second scrape intervals, with latency histograms across 10 buckets (5ms–5s) visualized on real-time Grafana dashboards.

DeployForge | Cloud-Native Internal Developer Platform & Deployment Engine | Repo
TypeScript, Node.js, BullMQ, Redis, PostgreSQL, Prisma, Docker, Trivy, AWS SDK (ECR/ECS/ALB), React

- Architected an event-driven Internal Developer Platform decoupling deployment requests from background execution via BullMQ and Redis, supporting concurrent builds with isolated filesystem sandboxing.
- Engineered a 10-state finite state machine orchestrating the microservice deployment lifecycle — Git checkout, multi-stage Docker builds, security scanning, and Blue/Green promotion — with a real-time dashboard streaming build logs and state transitions.
- Implemented an automated DevSecOps gate using Trivy CLI to scan container images for CVEs and secrets, halting pipeline execution on any critical finding.
- Integrated AWS SDK v3 (ECR, ECS, ALB) to automate container image publishing with SHA-256 digest auditing and Blue/Green traffic promotion via ALB listener re-weighting.

EDUCATION

Bachelor of Technology in Information Technology Sept 2023 – May 2027
Dr. A.P.J. Abdul Kalam Technical University (AKTU) | CGPA: 7.66

CERTIFICATIONS

AWS Certified Cloud Practitioner (CLF-C02) — Amazon Web Services | Issued Jul 2026 [↗]

ACHIEVEMENTS & POSITIONS OF RESPONSIBILITY

- Class Representative — Coordinated between faculty and 60+ students; resolved academic and administrative concerns in a timely manner.
- Student Coordinator, Startup Spotlight (Skillancer Pvt. Ltd.) — Led end-to-end event logistics, participant engagement, and on-ground execution. [Certificate]